This is a review submitted to Mathematical Reviews/MathSciNet.

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Title: Bayesian analysis with conditionally identically distributed sequences.

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Review text:

This paper develops Bayesian-style inference without assuming exchangeability, instead using the weaker *conditionally identically distributed* (c.i.d.) property. The key contribution is that prior and posterior distributions on a parameter space remain well-defined via c.i.d. sequences, with predictive distributions updated using copula-based martingales.

Let $X_{1:\infty} = (X_1, X_2, \dots)$ be a sequence of real-valued random variables. Exchangeability implies a random probability measure μ such that, given μ, observations are i.i.d. (de Finetti). The c.i.d. property relaxes this: for all $k > n \geq 1$ and bounded measurable g,

$$\mathbb{E}[g(X_k) \mid X_{1:n}] = \mathbb{E}[g(X_{n+1}) \mid X_{1:n}] \quad \text{a.s.}$$

A random probability measure μ still exists with

$$\mu(B) = \lim_{n\to\infty} \mathbb{P}(X_{n+1} \in B \mid X_{1:n}) \quad \text{a.s.}$$

A core novelty is constructing prior and posterior via the MLE. For a parametric model $f(x;\theta)$, define $\hat{\theta}_n = \arg\max_\theta \sum_{i=1}^n \log f(X_i;\theta)$. Under regularity conditions (strict concavity, integrability, tail bounds), Theorem 3.2 proves $\hat{\theta}_n \to \hat{\theta}_\infty$ almost surely. The distribution of $\hat{\theta}_\infty$ (starting from X_1) gives the prior. Given observed $X_{1:n}$, the tail sequence $X_{n+1:\infty}$ is c.i.d., and the distribution of $\hat{\theta}_\infty^{(n)}$ (from the tail) gives the posterior.

Copulas provide a constructive tool. Theorem 3.1: $X_{1:\infty}$ is c.i.d. iff there exist copulas $C_{x_{1:n}}$ such that

$$\mathbb{P}(X_{n+1} \leq x, X_{n+2} \leq y \mid X_{1:n} = x_{1:n}) = C_{x_{1:n}}\big(F(x \mid x_{1:n}), F(y \mid x_{1:n})\big),$$

where $F(x \mid x_{1:n}) = \mathbb{P}(X_{n+1} \leq x \mid X_{1:n} = x_{1:n})$. An iterative scheme is

$$F_{m+1}(x) = (1 - \alpha_m) F_m(x) + \alpha_m C_\rho(F_m(x), F_m(x_{m+1})),$$

with $\alpha_m = 1/m$ and Gaussian copula $C_\rho(u, v) = \Phi\big((\Phi^{-1}(u) - \rho\Phi^{-1}(v))/\sqrt{1-\rho^2}\big)$. This ensures F_m forms a martingale converging to μ.

A simpler parametric construction (Section 4.2) is

$$x_n = \theta_{n-1} + \sigma_{n-1} z_n, \quad \theta_n = (1 - a_n)\theta_{n-1} + a_n x_n, \quad \sigma_n^2 = \sigma_{n-1}^2 (1 - a_n^2),$$

with $z_n \overset{\text{i.i.d.}}{\sim} N(0, 1)$, $a_n \in (0, 1)$. Then $\theta_\infty = \lim \theta_n$ exists a.s. and defines the prior. For mixture models, only the sampled component updates, preserving c.i.d.

The paper provides almost-sure MLE consistency under c.i.d., extending classical i.i.d. theory. A subtle point raised by the reviewer is that the predictive distributions converge weakly to μ, but fast decay (e.g., $\sum \alpha_n < \infty$) ensures total variation convergence. This framework is valuable when exchangeability is restrictive or computationally intensive.